Physics Question Paper
Class 9
Holyfaith Academy, ICSE, Dhuliyan, MSD

Time: 2 Hours
Maximum Marks: 80

Section 1: Multiple Choice Questions (40 Marks)

Instructions: Choose the correct option for each question. Each question carries 2 marks.

(i) Which of the following quantities can be negative?
(a) Distance (b) Speed (c) Velocity (d) Both speed and velocity

(ii) A body has an acceleration $-10\,\mathrm{m\,s^{-2}}$. What is its retardation?
(a) $10\,\mathrm{m\,s^{-2}}$ (b) $-10\,\mathrm{m\,s^{-2}}$ (c) $-10\,\mathrm{m\,s^{-2}}$ (d) 2acc

(iii) Assertion (A): Thrust is a scalar quantity.
Reason (R): Thrust is force acting normally on a surface.
(a) Both A and R are true and R is correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true, R is false (d) A is false, R is true

(iv) Rocket works on the principle of Newton's:
(a) First law of motion (b) Second law of motion (c) Third law of motion (d) None of these

(v) If A and B are two objects with masses 10 kg and 20 kg respectively, then:
(a) A has more inertia than B (b) B has more inertia than A (c) B and A have the same inertia (d) None of the two has inertia

(vi) If the density of a liquid increases, the pressure at a point inside the liquid at a given depth:
(a) Increases (b) Decreases (c) Remains same (d) None of these

(vii) The hydraulic machines are based on:
(a) Newton's law (b) Boyle's law (c) Pascal's law (d) Newton's law

(viii) Sudden fall in the Mercury level indicates:
(a) The possibility of rain (b) Dry weather (c) Possibility of cyclone (d) Fair weather

(ix) If the pitch of the screw is 0.2 mm and there are 100 divisions on the circular scale, the least count of the given screw gauge is:

(a) 0.002 mm (b) 0.002 mm (c) 0.000 m (d) 0.2 mm

(x) In which direction does the buoyant force act on a body immersed in a liquid?
(a) Vertically upward (b) Sideways towards the walls of the container (c) Vertically downward (d) None of these

(xi) The standard atmospheric pressure is:
(a) 76 m of Hg (b) 76 cm Hg (c) 76 Pa (d) 76 mm of Hg

(xii) A straight line graph can be drawn between:
(a) L and T (b) L and T^2 (c) L^2 and T (d) L and $\sqrt{T}$

(xiii) From the given graph, the maximum velocity is:
Diagram placeholder: Insert velocity-time graph here. Please provide the image to extract the graph.

(xiv) Buoyant force depends on:
(a) Volume of body (b) Volume of displaced liquid (c) Density of body (d) Volume of liquid

Section 2: Descriptive Questions (40 Marks)

Instructions: Answer all questions. Marks are indicated against each.

Question 2 (8 Marks)
(i) Complete the following by choosing the correct answers from the brackets:
a. The instrument used to measure the diameter of a wire is _____ (Vernier scale/screw gauge/meter scale).
b. _____ (Distance/Displacement/Length) is the shortest distance between two points.
c. The slope of a displacement-time graph gives _____ (velocity/acceleration/speed).
d. There are _____ (one/two/three) types of inertia.
e. $1\,\mathrm{m\,s^{-1}} =$ _____.
f. The ratio of buoyant force experienced by a solid body immersed in two liquids whose relative density are 1 and 0.5 respectively is _____ (2:1/1:2/1:1).
(ii) Define uniform motion.
(iii) What kind of velocity is a flying bird most likely to have?

Question 3 (10 Marks)
(i) Derive the equation, $s = ut + \frac{1}{2}at^2$.
(ii) State Newton's second law of motion.
(iii) Calculate the acceleration of a body of 10 kg when a force of 50 N is applied on it.
(iv) Draw the acceleration-time graph for a body moving with:
(a) Uniform velocity
(b) Free-falling body and give an example of each.
Diagram placeholder: Insert acceleration-time graphs here. Please provide the image to extract the graphs.

Question 5 (10 Marks)
(i) (a) What is meant by fundamental units?
(b) Name three fundamental units.

(c) Name the two scales of a vernier caliper.
(ii) Derive the unit of the following in terms of standard fundamental units:
(a) Energy
(b) Pressure
(c) Density
(iii) A screw gauge has a least count of 0.001 cm and a zero error of 0.007 cm. Draw a neat diagram to represent it.
(b) Explain the terms pitch and least count of a screw gauge.
Diagram placeholder: Insert screw gauge diagram here. Please provide the image to extract the diagram.

Question 6 (6 Marks)
(i) Two bodies A and B of the same mass are moving with velocity V and $3V$ respectively. Compare their:
(a) Inertia
(b) Momentum
(c) The force needed to stop them at the same time
(ii) (a) Show that the rate of change of momentum = mass $\times$ acceleration.
(b) Under what condition does this relation hold?
(iii) Explain the following by giving a suitable reason:
(a) A passenger in a bus falls backward when it starts suddenly.
(b) An athlete runs a certain distance before taking a long jump.

Question 7 (6 Marks)
(i) (a) What do you understand by atmospheric pressure?
(b) What is the unit of atmospheric pressure?
(c) Write the advantage of using Mercury as a barometric liquid.
(ii) (a) Derive the expression $P = h\rho g$ (where P is pressure, h is height, ρ is density of liquid, g is acceleration due to gravity).
(b) Calculate the water pressure at a depth of 10 m, with a density of water of $1000\,\mathrm{kg\,m^{-3}}$.

Question 8 (6 Marks)
(i) (a) State Pascal's law.
(b) Calculate the output load from a big piston of a hydraulic jack when a 10 N force is applied on a small piston and the ratio of the cross-section area of their piston is 1:10.
(c) Is Pascal's law applicable for open containers?
(ii) (a) Why does one's nose start bleeding on high mountains?
(b) A ball is moving on a perfectly smooth horizontal surface. If no force is applied on it, will its speed decrease, increase, or remain unchanged?
(iii) The velocity-time graph of a moving body is given below.
Find:
(a) The acceleration in parts AB, BC, and CD.
(b) The displacement in each part AB, BC, CD.
(c) The total displacement.
Diagram placeholder: Insert velocity-time graph here. Please provide the image to extract the graph.